



Basic Science of Implants in Sports Medicine

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Sports injuries in athletes are commonly associated with injury to soft tissues, specifically tendons, ligaments, menisci, and cartilage. Common tendon and ligament injuries in the physically active population include rotator cuff tears, Achilles tendinopathy, anterior cruciate ligament (ACL) tears, and lateral epicondylitis (“tennis elbow”). These injuries can be classified as either repetitive microtrauma, which is caused by overuse, or macrotrauma, which typically results from external physical impact.¹ Because each joint is a complex system that is made up of different types of tissue that work together, trauma or degeneration of one tissue often results in the injury to another tissue. For example, in the knee, a common sports-related injury such as an ACL tear is usually associated with subsequent damage to the meniscus and/or the neighboring articular cartilage. This in turn leads to a longer-term increased risk for osteoarthritis. Thus an understanding of tendons, ligaments, menisci, cartilage, and other joint soft tissues is essential for the clinical management of sports injuries.

Current options used to treat sports injuries in orthopedics include autografts, allografts, and tissue-engineered grafts. Autografts are harvested from the patient and remain the “gold standard” for reconstruction. However, they also have disadvantages, such as longer surgical time due to graft acquisition, along with the risk of donor-site morbidity. Allografts eliminate the need to harvest grafts on site and the risk of donor-site morbidity. Although they have been used successfully for musculoskeletal tissue reconstruction, they are not appropriate for every situation and have some limitations.^{2–7} The major shortcomings of allografts are their limited availability, a lack of functional integration with the surrounding host tissue, and, although low, the potential risk of disease transmission or an immunologic response.⁸

To overcome the limitations of biologic grafts, synthetic grafts have been developed as an alternative to autografts and allografts for orthopedic applications. Because these implants are fabricated out of synthetic materials, naturally derived materials, or a combination of the two, the risk of disease transmission is either nonexistent (in the case of synthetics) or significantly reduced (in the case of naturally derived materials). For connective tissue repair, polymers (e.g., polyethylene, polylactic acid [PLA], and polycaprolactone) are commonly used to form the implants, many of which are already used in medical devices approved by the US Food and Drug Administration (FDA).^{9–11} Examples of

naturally derived biopolymers include collagen, chitosan, and alginate.^{12–15} These materials have been tested for biocompatibility using in vitro culture models or in vivo animals models to reduce the risk of adverse responses when used clinically.^{16–18} The versatility and sophistication of their properties can be fine-tuned by manipulating the materials to improve biocompatibility, bioactivity, mechanical properties, and integration with the host tissue.^{19–23}

In addition to synthetic grafts, recent advances related to tissue engineering have promoted the development of cell-based and scaffold-based approaches to orthopedic tissue regeneration. Unlike permanent implants, scaffold systems work as temporary structures supporting tissue formation by cells. Early synthetic scaffold systems relied on host cells to infiltrate the scaffold at the repair site; in more advanced scaffolds, biologic molecules are preincorporated and/or cells are preseeded onto the scaffolds prior to implantation. For both cases the scaffold degrades away and the structural template is replaced completely by new tissue, which is achieved by maintaining a delicate balance in cell biosynthesis and scaffold degradation. Thus when designing these implants, the materials and implant morphology need to elicit a cell response favorable for formation of the site-specific tissue. The mechanical properties also need to match the native tissue to support loading while the cells deposit new tissue, and the scaffold needs to be designed to degrade at an appropriate rate. Currently many implants have been approved by the FDA for tendon augmentation and cartilage repair, with many more devices undergoing clinical trials and further research.

This chapter focuses on synthetic implants for the treatment of tendon, ligament, meniscus, and cartilage injuries, highlighting those that have either been approved by the FDA or have reached the clinical trial stage. For cases in which synthetic grafts are not yet available, tissue-engineered implants are discussed. Each section begins with a brief overview of the tissue of interest, followed by a discussion of current synthetic or tissue-engineered grafts. A brief summary is included at the end of each section.

IMPLANTS FOR TENDON REPAIR

The rotator cuff consists of four tendons that attach their respective muscles to the proximal humerus through direct

Abstract

This chapter focuses on synthetic implants for the treatment of tendon, ligament, meniscus, and cartilage injuries, highlighting those that have either been approved by the US Food and Drug Administration or have reached the clinical trial stage. For cases in which synthetic grafts are not yet available, tissue-engineered implants are discussed. Each section begins with a brief overview of the tissue of interest, followed by a discussion of current synthetic or tissue-engineered grafts. A brief summary is included at the end of each section.

Keywords

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tendon-to-bone insertions.²⁴ The rotator cuff acts to stabilize the glenohumeral joint and is prone to degeneration and injury, with cuff tears being the most common form of pathology afflicting the shoulder. Each year more than 250,000 rotator cuff repairs are performed in the United States alone.²⁵ The incidence of rotator cuff repair is also increasing because of an aging, yet active, population. Primary tendon-to-bone repair and healing are the goals of these surgical procedures. In some cases, augmentation with commercially available patches is performed, with the patch applied to the superficial tendon surface to enhance the repair. However, because of a variety of factors including graft degeneration, poor vascularization, muscle atrophy, and the lack of graft-to-bone integration after surgery,^{26–30} failure rates between 20% and 94% have been reported after primary repair of chronic rotator cuff injuries.³¹

To improve healing, biologic or synthetic polymer-based tendon implants or augmentation devices have been developed to reconstruct large and massive rotator cuff tears.^{32,33} To date, most biologic tendon grafts that are commonly used are derived from a decellularized allogeneic or xenogenic extracellular matrix (ECM).^{34–37} Synthetic grafts are usually made of biocompatible and biodegradable polymers that break down into nontoxic metabolites in the body. There are several commercially available tendon grafts currently available, as well as several ongoing clinical trials testing new technology for these implants.

Most commercially available synthetic grafts are developed around biologic materials derived from ECM, such as small intestinal submucosa (SIS) and dermis. These patches provide a chemical and three-dimensional (3D) structural framework, native matrix composition, and residual remodeling biomolecules that direct repair and remodeling of the rotator cuff tendons by the host cells.³⁸ However, their clinical use, especially for SIS-based grafts, has been in question due to suboptimal or negative outcomes in human trials.^{39,40} Several reported adverse outcomes have been reported, attributed either to immunologic issues, and/or a mismatch in mechanical properties between the graft and the host tissue; this has been particularly apparent in the demanding environment of the shoulder joint. A systematic comparison of four commercially available ECM patches (Restore, derived from porcine SIS; CuffPatch, derived from porcine SIS; GraftJacket, derived from human dermis; and TissueMend, derived from bovine dermis) was conducted using a canine model.⁴¹ All four patches were inferior mechanically to the native tendon and underwent premature graft resorption.

The ineffective results using natural grafts motivate the search for appropriate synthetic grafts for tendon repair. However, synthetic grafts may also incite a local negative response due to local toxicity or acidity from graft degradation products. Examination of synthetic patches in canine and rat models^{42,43} showed that both patches (X-Repair, made of poly-L-lactide, and Biomerix RCR Patch, made of polycarbonate polyurethane) were biocompatible with host cell responses and showed minimal inflammation response, supporting their use for tendon repair. Alternatively, hybrid rotator cuff patches are also available, synthesized with the aim of combining the benefits of both poly-L-lactide and polyurethane. However, very limited data are available for the performance of these scaffolds. Currently, Tornier BioFiber-CM,

a patch made by adding bovine collagen I to poly-4-hydroxybutyrate, is in clinical trials for full-thickness rotator cuff tear repair.⁴⁴

Although rotator cuff patches are frequently used in surgery, systematic follow-up studies that evaluate performance of such scaffolds are sparse. Results from the limited number of follow-up studies available demonstrated a mixed performance of commercially available patches. Restore, an SIS-based scaffold, was the first implant on the market for tendon repair and was widely used. However, clinical outcomes for Restore have been mixed. An early study reported that, when compared with preoperative levels, shoulders repaired with the Restore patch improved in strength, motion, and function, with no increased risk of infection at 24 months.⁴⁵ Sclamberg et al.⁴⁰ performed a 6-month follow-up study of 11 patients (5 women and 6 men between the ages of 52 and 78 years) who underwent large rotator cuff repair augmented with the Restore implant and found that the repair failed in 10 of the 11 patients. Of even greater concern were the results reported by Iannotti et al.³⁹: in a study of 15 patients (4 women and 11 men with mean age of 58 years), those treated with the Restore patch had a lower healing rate and a lower postoperative PENN score compared with members of the control group, who underwent repair without the patch. In a later study, Walton et al.⁴⁶ compared healing results of patients (5 women and 10 men with a mean age of 60.2 years) who had their rotator cuff repaired with or without the Restore implant.⁴⁶ It was observed that 2 years after surgery, patients who received the Restore implant reported higher rates of repeat tears, and their measured mechanical properties were significantly weaker. The patients also had more impingement and decreased levels of sports participation. In another study, Malcarney et al.⁴⁷ examined the postoperative responses of 25 patients who underwent rotator cuff repair augmented with the Restore implant and found that four of them experienced an acute inflammatory response shortly after surgery (a mean of 13 days). The inflammatory responses were nonspecific, and all four patients required implant removal. Based on these results, SIS-based scaffolds such as Restore are not recommended for use in rotator cuff repair.

The next generation of ECM-based implants focused on improving matrix mechanical properties to more closely match the mechanical properties of tendon. TissueMend is derived from fetal bovine dermis and is decellularized through a series of chemical processes. Magnussen et al.⁴⁸ used TissueMend to repair cadaveric Achilles tendons and tested the tendon mechanical properties under cyclic tensile loading. Results showed decreased gapping in the implant-augmented group, and the ultimate failure load increased significantly from a mean of 392 N to 821 N for the control group compared with the augmented group, respectively. Although currently the literature does not include any reports on the use of the TissueMend implant for rotator cuff repair, Seldes and Abramchayev⁴⁹ demonstrated, in a cadaver, the feasibility of using this implant to repair a massive rotator cuff tear.

Another implant available for rotator cuff repair is the Zimmer Collagen Repair Patch, which is derived from porcine dermis. In addition to being decellularized, this patch is cross-linked with hexamethylene diisocyanate to increase its strength and

stability. The Zimmer Collagen Repair Patch also similarly showed inconsistent performance. Proper et al.⁵⁰ used the Zimmer patch to repair massive rotator cuff tears in 10 patients (5 women and 5 men between the ages of 46 and 80 years) and found that the implant caused no major postoperative complications and no adverse reactions. These investigators also found that the pain score and interval function score improved at 1 year after surgery; similarly, better range of movement and strength were reported. In the medium-term follow-up (a duration of 3 to 5 years), Badhe et al.⁵¹ found that although two implants had detached after the original surgery, pain was reduced and abduction power and range of motion improved significantly, with no adverse effects reported. However, when Soler et al.²⁶⁵ used the Zimmer patch to repair massive rotator cuff tears in four patients (three women and one man between the ages of 71 and 82 years), all four grafts failed between 3 and 6 months after surgery, leading to recurrent tears.

Compared with the SIS-based scaffold and Zimmer Collagen Repair Patch, published results evaluating GraftJacket have demonstrated more consistent results. In a prospective, randomized study, patients with a massive rotator cuff tear repaired with GraftJacket showed improved pain scores and a higher ratio of intact tendon at the 24-month follow-up compared with the patients with shoulders repaired without the graft.⁵² In other studies, results demonstrated that augmentation with GraftJacket led to a lower retear rate, improved pain score, and increased shoulder functionality compared with the preoperative condition with no inflammatory response.^{53–55} Currently, only one study is available reporting the performance of the synthetic Biomerix RCR Patch. In this study, patients showed improved pain scores, satisfactory range of shoulder movement at 6 and 12 months, a low retear rate (10%), and no adverse reactions.⁵⁶

Barber et al. evaluated the mechanical properties of major commercial implants (Restore, Permacol, TissueMend, and GraftJacket) under tension.⁵⁷ These investigators found that products derived from human skin were the strongest, followed by porcine and bovine skins, with the small intestine submucosa-based implant being the weakest. This difference in mechanical properties could be the potential cause of the difference in performance of the implants in patients. Even with the limited number of follow-up studies performed, it is clear that despite a wide selection of commercial patches, very limited success has been found in early clinical trials. Surgical outcomes are also associated with other nonpatch factors such as age of the patient, size and severity of the tear, and surgical techniques used. Surgeons should therefore keep these factors in mind when evaluating the literature and be cautious when selecting an augmentation graft.

Because of the aforementioned challenges associated with ECM-derived implants for rotator cuff repair, a demand exists for new technologies to better meet the needs of functional tendon regeneration and soft tissue-to-bone integration. For integrative rotator cuff repair, the scaffold should match the mechanical properties of the native tendon, which is the major limitation of available biologic implants. The ideal implant should also mimic the ultrastructural organization of the native tendon. In addition, the implant should be biodegradable so it can be gradually replaced by new tissue while maintaining its physiologically

relevant mechanical properties. Lastly, the graft must integrate with the host tendon and surrounding bone tissue by promoting the regeneration of the native tendon-to-bone interface. Several groups have explored synthetic grafts and tissue engineering methods for the development of tendon implants.^{1,58–60} It is common to use scaffolds composed of nanofibers based on a variety of synthetic polymers, such as poly-L-lactic acid (PLLA), polylactide-co-glycolic acid (PLGA), polycarbonate polyurethane, and biologic materials, such as collagen and silk.^{1,56,60–63} In addition to being biodegradable, both PLLA and PLGA are materials already used in FDA-approved devices. In a study by Baker et al.⁶⁴ using a canine model, 11 subjects underwent bilateral rotator cuff repair with one shoulder using a novel PLLA (human) fascia patch and then other without to assess whether these patches enhance the strength or likelihood of healing of the repair. At time 0 and at 12 weeks, the group compared repair retraction, cross-sectional area, biomechanical properties, and biocompatibility between the two sides. They found that at time 0 the patch side was able to withstand higher loads ($296 \text{ N} \pm 130 \text{ N}$ more) than the contralateral side; however, at 12 weeks, the augmented sides could withstand less ($192 \text{ N} \pm 213 \text{ N}$) than the nonaugmented side and there was no difference in stiffness between the groups.⁶⁴ McCarron et al.⁶⁵ looked at cyclic gap formation and failure properties of a scaffold-augmented rotator cuff repair in human cadaveric shoulders. In nine paired cadaveric shoulders, the augmented and nonaugmented controls were loaded from 5 N to 180 N for 1000 cycles. They found that the augmented shoulder had a statistically significantly decreased gap at cycles 1, 10, 100, and 1000 compared with the nonaugmented repairs. Notably, all of the augmented repairs successfully completed the full cycling trials, whereas three of the nine nonaugmented repairs failed before completion. The gap remained less than 5 mm for the augmented group, whereas the nonaugmented group averaged a 7.3-mm gap. These studies are part of the ongoing body of work that will expand to include additional animal and eventually human studies to fully determine the efficacy of fascially augmented rotator cuff repairs.

The nanoscale architecture of the collagen-rich tendon matrix can be readily recapitulated with nanofiber scaffolds, which exhibit high surface area-to-volume ratio, low density, high porosity, variable pore size, and mechanical properties approximating those of the native tissues. Moffat et al.²¹ first reported on the fabrication of PLGA nanofiber scaffolds with physiologically relevant structural and mechanical properties for rotator cuff repair. It was observed that human rotator cuff fibroblast morphology and growth on aligned and unaligned fiber matrices were dictated by fiber alignment, with distinct cell morphology and integrin expression profiles. Upregulation of $\alpha 2$ integrin, a key mediator of cellular attachment to collagenous matrices, was observed when the fibroblasts were cultured on aligned fibers, and upon which types I and III collagen-rich matrices were deposited. More recently, Xie et al.⁶⁶ developed a continuous PLGA nanofiber scaffold that transitioned from aligned to random orientation, to examine the effects of this transitional region on rat tendon fibroblasts in vitro. After 1 week of culture, the cells proliferated on both aligned and random nanofiber orientations. Although a rounded morphology was found on unaligned nanofibers, cells

cultured on aligned nanofibers appeared long and spindle-like and were aligned along the long axes of the fibers.

The biologic response to polymeric nanofibers may also be enhanced by additional surface modifications. For example, Rho et al.⁶⁷ electrospun aligned type I collagen nanofiber scaffolds with a mean fiber diameter of 460 nm and evaluated the response of human epidermal cells after coating the scaffolds with several adhesion proteins. It was found that cell proliferation was enhanced by coating the scaffolds with both type I collagen and laminin. Park et al.⁶⁸ applied a plasma treatment to polyglycolic acid (PGA), PLGA, and PLLA nanofibers and grafted a surface layer of hydrophilic acrylic on these scaffolds. It was found that NIH-3T3 fibroblasts seeded on these modified scaffolds spread and proliferated faster than those on unmodified control scaffolds. Nanofibers have also been used to improve existing scaffold design, resulting in a graft with a more biomimetic surface for eliciting desired cell responses. For example, Sahoo et al.⁶¹ electrospun PLGA nanofibers directly onto a woven microfiber PLGA scaffold to increase cell seeding efficiency while maintaining a scaffold that was mechanically competent. The attachment, proliferation, and differentiation of porcine bone marrow stromal cells was evaluated on these scaffolds and, when compared with scaffolds seeded via a fibrin gel delivery, it was found that seeding the cells onto nanofiber-coated scaffolds enhanced proliferation and collagen production and upregulated the gene expression of several tendon-related markers, namely decorin, biglycan, and type I collagen.

In addition to being used as synthetic tissue-engineered implants, polymer fibers can be used to modify allogeneic non-tendon tissue and make it more suitable for rotator cuff augmentation. Aurora et al.⁶⁹ sutured PLLA and PLLA/PGA braids (diameter = 400 μ m) to a human fascia patch to increase its suture retention properties. Results showed that all reinforced patches withstood 2500 cycles of 5 N to 150 N cyclic loading, with the PLLA/PGA patch withstanding 5000 cycles of loading. In addition, the suture retention properties and the maximum construct load of the reinforced patch were observed to be greater than those of human rotator cuff tendon, even after 3 months in vivo. Although more foreign body giant cells were seen in the reinforced patches, it is expected that this response will decrease after the polymer has fully degraded. These promising results suggest that the polymer fiber-reinforced tendon patch could be a more functional alternative for rotator cuff augmentation.

To address the challenge of regenerating the tendon-to-bone insertion site, several groups have evaluated the feasibility of integrating tendon implants with bone or biomaterials through the formation of anatomic insertion sites. Fujioka et al.⁷⁰ examined the effects of reattaching the bone and tendon in a rat model for Achilles tendon avulsion. After 4 weeks, surgical reattachment of tendon to bone increased type X collagen deposition and allowed tissue to maintain distinct regions of calcified and noncalcified fibrocartilage tissue. In addition, Inoue et al.⁷¹ promoted supraspinatus tendon integration with a metallic implant using a bone marrow-infused bone graft. These early attempts demonstrate that the direct tendon-bone insertion may be regenerated. To this end, the ideal implant for tendon-to-bone interface tissue engineering must exhibit a gradient of structural and

mechanical properties mimicking those of the multitissue insertion.⁷² Thus a scaffold recapturing the nanoscale interface organization, with preferentially aligned nanofiber organization and region-dependent changes in mineral content, would be highly advantageous. Building on the functional PLGA nanofiber scaffold designed for tendon tissue engineering, Moffat et al.^{73,74} designed a biphasic scaffold, with the top layer consisting of nanofibers of PLGA and the second layer consisting of composite nanofibers of PLGA and hydroxyapatite nanoparticles. The biphasic design is aimed at regenerating both the nonmineralized and mineralized fibrocartilage regions of the tendon-to-bone insertion site while promoting osteointegration with PLGA-HA nanofibers.⁷³ The responses of tendon fibroblasts, osteoblasts, and chondrocytes have been evaluated on these nanocomposite scaffolds, with promising results in vitro. When tested in vivo subcutaneously, as well as in a rat rotator cuff repair model, the biphasic scaffold supported regeneration of continuous noncalcified and calcified fibrocartilage regions, demonstrating the potential of a biodegradable nanofiber-based implant system for integrative tendon-to-bone repair.^{73,74} Lastly, the efficacy of the biphasic scaffold to repair acute, full-thickness rotator cuff tears was confirmed in a sheep model. In an alternative approach, nanofiber PLGA scaffolds with gradients in mineral content were produced using electrospinning.^{75–77} The gradients in mineral content resulted in gradients in mechanical properties and cellular responses, mimicking features of the natural tendon-to-bone attachment. These scaffolds showed some promise in a small animal model of rotator cuff repair.⁷⁸ Collectively, these results demonstrate the potential of the biomimetic, biphasic scaffold for integrative, tendon-bone repair.

In summary, commercially available, ECM-derived tendon grafts and augmentation devices have exhibited mixed outcomes in clinical studies. In general, tendon implants made from dermis performed better than implants made from other types of ECM, as reflected in the reported lower repeat tear rates and improved postoperative scores. Alternative treatment options such as synthetic and tissue-engineered grafts are being developed, with promising results for tendon regeneration and tendon-to-bone integration. However, several challenges remain to be overcome before the widespread clinical application of the tissue-engineered tendon implants can be realized. For example, one challenge is the scale-up of the tissue-engineered implants from small animal models to humans. Currently, most of the tissue-engineered implants are being evaluated in vitro and are prepared in small batches. High-throughput fabrication and delivery processes need to be developed for them to augment their commercial applicability. The other challenge is that the fabrication process of the tissue-engineered implants generally uses a variety of toxic solvents or several steps of chemical reactions to dissolve polymers, which may have undesired effects on biomolecules or cells once they are implanted in humans.

IMPLANTS FOR LIGAMENT RECONSTRUCTION

The ACL is one of the major ligaments that connect the femur and tibia, and it is the primary stabilizer in the knee. The ACL inserts into bone through a direct transition consisting of spatial

variations in cell type and matrix composition, resulting in three distinct tissue regions of ligament, fibrocartilage, and bone.^{79–81} Tearing of the ACL is among the most common knee injury afflicting the young and active population, with more than 100,000 ACL reconstructions performed annually in the United States alone.^{82,83} Because of the inherent poor healing potential of the ligament, ACL reconstruction based largely on biologic grafts is required. Because of the relative scarcity and donor site morbidity of autografts, as well as the inherent risks associated with allografts, significant interest has been expressed regarding synthetic- and tissue-engineered alternatives for ACL reconstruction. In the late 1980s the FDA approved synthetic materials for use as an alternative implant for ACL reconstruction. However, because of significant complications and failures in humans, all of the synthetic implants were withdrawn from the market by the late 1990s, and at the present, only biologic grafts such as hamstring tendon or bone-patellar tendon-bone grafts are used clinically for ACL reconstruction.

One of the commercial implants tested for ACL reconstruction was the Gore-Tex implant, which is based on polytetrafluoroethylene (PTFE). This implant is composed of solid PTFE nodes interconnected by PTFE fibers at each end for bony attachment.⁸⁴ In 1987 Ahlfeld et al.⁸⁵ used the Gore-Tex implant to treat 30 patients with unstable knees. During the follow-up period (average 24 months), the group treated with the Gore-Tex implant showed improved satisfaction compared with patients who underwent reconstruction with a different material (ProPlast ligament) (83% satisfactory vs. 52%). Glousman et al.⁸⁶ used this implant in 82 patients (23 women and 59 men, ages 16 to 51 years) and at 18 months follow-up found that a number of patients had complications (15) and repeat operations (14). On the other hand, the subjective scores improved in all evaluation categories, including pain, swelling, giving way, locking, and stair climbing. Although the authors suggested that the results were positive, they also recommended longer-term follow-up before definitive conclusions could be determined. In another study, Indelicato et al.⁸⁷ reported that of 39 patients (12 women and 27 men whose ages ranged between 17 and 42 years) who received the Gore-Tex implant, 87% had satisfactory results 2 years post-operatively. However, in many of these studies, some complications were noticed, such as tears of the implant and sterile effusions. In addition to the failure of the implant itself, use of Gore-Tex led to other complications in the knee, including bone tunnel widening. Muren et al.⁸⁸ examined 17 patients at 13 to 15 years after they had ACL reconstruction with the Gore-Tex implant. It was found that six patients underwent revision surgery as a result of implant rupture and pain; moreover, 15 patients had tibia bone tunnel widening. In another study, surgery with the Gore-Tex implant was performed in 123 patients, and complete rupture of the graft was seen in 26 patients.⁸⁹ In addition, half of the patients exhibited graft loosening, 62% experienced osteoarthritic change, and bone tunnel osteolysis at both ends was identified in most cases. Consequently, the Gore-Tex implant was no longer recommended for ACL reconstruction and eventually was withdrawn from the market in 1993. Similarly, for other synthetic ACL implants such as the Ligament Advanced Reinforcement System (LARS; Surgical Implants and Devices,

Arc-sur-Tille, France) and Leeds-Keio ligament implants that were used in ACL reconstruction with short-term satisfactory results, many long-term complications were reported, including repeat rupture, bone tunnel widening, severe synovitis, and inflammatory responses. Therefore these implants were likewise withdrawn from the market, and currently no synthetic implants have been approved by the FDA for ACL reconstructions.

To overcome limitations of the failed synthetic implants and the inherent shortcoming of allografts currently in use, tissue-engineered implants have been investigated. Such implants involve implantation of a bioactive material that regenerates tissues with material properties that are comparable with those of the native ACL.⁹⁰ Similar to the requirement for tendon repair, the ACL implant should be biodegradable, match the mechanical properties of the native ACL, and mimic the ultrastructural organization of the native ACL. Finally, the implant must integrate with both the femoral and tibial bone tunnels to promote the regeneration of the native ligament-to-bone interface. To this end, most of the studies have investigated the use of synthetic polymers such as PLLA, PLGA, and polyurethane, as well as biologic materials, such as collagen and silk.^{91–95}

Dunn et al.⁹⁶ were among the first investigators to evaluate a biomimetic ACL replacement *in vivo*. Studies were performed using a type I collagen fiber-based prosthesis with polymethylmethacrylate (PMMA) bone fixation plugs on the ends. Although neoligamentous tissue formation was reported, the majority of scaffolds were reported to have ruptured after 20 weeks in a rabbit model, demonstrating that, although it was biomimetic, this system was insufficient to support long-term knee stability.⁹⁶ A series of studies also were performed that focused on the development of a silk-based ACL replacement both *in vitro* and *in vivo*.^{94,95,97–99} Specifically, a novel silk-fiber based scaffold was designed, and several studies were performed to assess the impact of chemical and mechanical stimulation on the differentiation of seeded mesenchymal stem cells (MSCs). It was demonstrated that individually, soluble chemical factors such as basic fibroblast growth factor, as well as tensile and torsional loading, could drive matrix elaboration and fibroblastic differentiation of MSCs on the silk scaffold.⁹⁴ In addition, chemical and mechanical stimulation, when applied concomitantly and controlled temporally, have been reported to enhance MSC response.^{97,98} The system was implanted *in vivo* in a goat model, with promising results reported based on histologic and mechanical outcomes such as significantly increased knee stiffness and ultimate tensile strength after 12 months of implantation. This prosthesis is currently undergoing clinical trials.⁹⁹

Also using a silk-based scaffold, Liu et al.¹⁰⁰ performed a series of studies to optimize a knitted graft combined with a micro-porous silk sponge for ACL reconstruction. This biphasic system was designed to mimic the ECM of native tissue and provide sufficient mechanical strength for ligament replacement. The scaffold was implanted *in vivo* using a rabbit model and later a pig model, with substantial ligament-like tissue observed on the scaffold after 24 weeks.^{101,102} In addition, several *in vitro* studies were performed to further optimize the scaffold, including the use of silk cables to increase tensile strength, the incorporation of basic fibroblast growth factor-releasing PLGA and RADA16

peptide nanofibers to enhance cell biosynthesis, and the addition of a silk-based aligned nanofiber topography to direct cell orientation and matrix production.^{62,103–105}

Progressing from single-phase systems, Cooper et al.^{91,92,106} designed and optimized, both in vitro and in vivo, a braided, α -hydroxyester, microfiber-based scaffold for ligament engineering. The architecture and porosity was standardized in vitro using a braiding technique to recapitulate the native ligament mechanical properties and included a denser fiber arrangement at each end of the construct for bone formation.¹⁰⁶ Scaffold composition was also optimized based on in vitro degradation and cell response, with PLLA selected because of its ability to maintain structural integrity over the 8-week culturing duration.⁹² Subsequently, the optimized scaffold was evaluated in vivo using a rabbit ACL reconstruction model. It was demonstrated that cell seeding of the implanted scaffold resulted in a marked improvement in functional outcomes, but scaffolds in both groups ruptured after 12 weeks of implantation.⁹¹ In a follow-up study, Freeman et al.^{107,108} evaluated the effect of both braiding and twisting on the mechanical properties of the PLLA microfiber system, demonstrating that twisting coupled with braiding could increase both ultimate tensile strength and the toe-region length of the graft. Barber et al.¹⁰⁹ reported on the development of a braided nanofiber-based scaffold for ACL replacement. Scaffold architecture was optimized by varying the number of braided bundles and evaluating mechanical properties, with minimal differences in toe-region or elastic modulus as a function of braid number. The scaffold was seeded with MSCs in vitro, and both cell viability and proliferation were observed.

Also in development are scaffold systems that target the regeneration of the ligament-to-bone interface, which is critical for biologic fixation of either biologic or synthetic grafts. Spalazzi et al.^{110,111} reported on the design and evaluation of a triphasic scaffold for the regeneration of the ACL-to-bone interface both in vitro and in vivo. The scaffold consists of three distinct yet continuous phases, each engineered for a specific tissue region found at the interface: Phase A is designed with a PLGA mesh, phase B consists of PLGA microspheres, and phase C is composed of a sintered PLGA and 45S5 bioactive glass composite.¹¹² They are intended for ligament, interface, and bone formation, respectively. When this stratified scaffold was evaluated in a subcutaneous athymic rat model, abundant tissue formation was observed on phases A, B, and C. Cell migration and an increased matrix production were also observed in the interfacial region, and the phase-specific controlled matrix heterogeneity was maintained in vivo. Once ligament fibroblast, chondrocyte, and osteoblast triculture were established on their respective phase of the scaffold, the formation of both anatomic ligament-like and bonelike matrices was observed on the triphasic scaffold (phases A and C, respectively), as well as the deposition of a fibrocartilage-like tissue (phase B). At 2 months after implantation, the interface-like region consisted of chondrocyte-like cells embedded within a matrix containing collagen types I and II, as well as glycosaminoglycans, indicating the formation of interface-like tissue.

In addition to ACL reconstruction, a new approach emphasizing primary ACL repair has been developed in response to the issue that ACL reconstruction was not delaying premature onset

of osteoarthritis in patients with ACL injury.^{113–115} Through a series of in vitro and in vivo experiments, a proprietary scaffold that combines collagen-based implants with whole blood for ACL repair has been designed and optimized.¹¹⁶ This system was first evaluated in an ACL central defect model, in which the collagen implant was augmented with platelet-rich plasma (PRP) was used only to fill in the defects, not to replace the ACL. In a canine model, Murray et al.^{116,117} used collagen with or without PRP to repair an ACL central defect and evaluated histologic and mechanical properties of the repaired ACL over a period of 6 weeks. Results indicate that the collagen gel with PRP showed a significantly higher percentage of defect filling and strength compared with the control group. In addition, it was found that the PRP-augmented collagen resulted in regenerated tissue that had similar properties to those of the medial collateral ligament, which has a better healing ability than the ACL. This collagen-platelet composite (CPC) is also used as a supplement to the standard allograft reconstruction approach. In one study, Joshi et al.¹¹⁸ performed unilateral ACL reconstruction procedures in pigs with a bone-patellar tendon-bone allograft with or without the addition of CPC at the surgery site, and outcomes were evaluated at 4 weeks, 6 weeks, and 3 months after surgery. Although at 6 weeks a temporary decrease in yield load and stiffness was seen in both groups, by 3 months the group that underwent repair with CPC had improvements in yield load and linear stiffness. In another study, using the same animal model, Fleming et al.¹¹⁸ evaluated the effect of the addition of CPC over a period of 15 weeks. Results confirmed the previous findings that in the long term, yield and maximum failure load of CPC-supplemented groups were greater than that of the group that underwent standard ACL reconstruction. In addition, histologic analysis revealed that the graft structure properties were also improved by the addition of CPC. Furthermore, the composite can be used by itself as an implant to enhance the suture repair procedure on the ACL. Two studies were conducted to evaluate the individual effects of collagen or PRP in ACL regeneration in a porcine model, and it was found that neither improved the functional properties.^{119,120} Mastrangelo et al.¹²¹ combined collagen with PRP and used CPC with different PRP concentrations ($\times 3$ and $\times 5$ of the baseline, respectively) to bridge the ACL stump and the femoral tunnel and evaluated the mechanical properties of the repaired ACL over 13 weeks. It was found that regardless of PRP concentration, CPC increased the mechanical properties of repaired ACL. Further in vitro work found that both platelets and plasma proteins were important in the healing process of the ACL cells, red blood cells aided with collagen production by fibroblasts, and white blood cells released anabolic growth factors.^{122–127} This led to the determination that PRP was no better than whole blood in the augmentation of the collagen scaffold for enhanced ACL repair.^{128,129} Therefore the ongoing trials testing the collagen scaffold in large animal models and humans used whole blood. In a large animal model, results at 6 and 12 months showed that ACL repair with the collagen scaffold and whole blood augment had similar mechanical properties to ACL reconstruction.^{130,131} Importantly, they also noted that the porcine knee that underwent ACL repair with the collagen scaffold and whole blood had significantly less osteoarthritis

at 1 year than the knee that underwent ACL reconstruction.¹³⁰ Based on these promising animal model studies, a first in human phase I FDA-approved trial of 20 patients was performed.¹³² Ten patients underwent ACL repair with the collagen scaffold, whole blood, and a primary suture repair of the ACL, and 10 patients underwent reconstruction with a hamstring autograft. There were no significant inflammation or joint injections in either group, no difference in pain or effusion, and MRI at 3 months post-operative showed intact ACL or graft in all patients. Hamstring strength was significantly stronger in the ACL repair group.¹³²

Murray et al.¹³³ is currently conducting an FDA-approved randomized controlled trial on 100 patients using the proprietary collagen scaffold to enhance repair of proximal femoral avulsion ACL injuries with promising early results.

In summary, because of the inherent poor healing potential of ligaments such as the ACL, implants are used for ligament reconstruction, with autografts and allografts being the most common. In the 1980s, synthetic ACL such as Gore-Tex, LARS ligament, and Leeds-Keio were popular and were used widely in ACL reconstruction surgeries. Although satisfactory short-term results were reported, long-term outcomes were poor and eventually resulted in most of the synthetic implants being extracted and withdrawn from the market. To address the unmet market demand for ligament grafts that were an alternative to biologic grafts, tissue engineering has arisen as a promising method by which to regenerate the ACL. A variety of polymeric materials were tested in vitro and in vivo; different methods such as incorporation of growth factors and active loading of the scaffolds were used to enhance graft performance. In addition, stratified implants were designed and showed promising results in vitro and in vivo for ligament-to-bone integration. Although they are promising, most of the tissue engineering options are still at the in vitro and small animal in vivo evaluation stages, and their true clinical potential remains to be demonstrated in clinical trials. One recent approach in clinical trials that uses natural scaffolding combined with whole blood may allow for primary ACL healing and has the potential to shift the care of these injuries from reconstruction to repair, which may be instrumental in delaying the onset of osteoarthritis after ACL injury.

IMPLANTS FOR MENISCUS REPAIR

The meniscus is a fibrocartilaginous tissue in the knee that functions to dissipate compressive and shear stresses during normal activity, as well as to distribute synovial fluid throughout the knee during loading and unloading.^{134,135} With a high water content (78 weight percent), the meniscus is 80% avascular and is composed primarily of type I and II collagen, with other minor types of collagen, as well as proteoglycans.¹³⁶ Meniscal tears can either be traumatic or degenerative in nature. The current treatment options include nonoperative (e.g., physical therapy, nonsteroidal antiinflammatory medications, and cortisone injections) and operative procedures, which include meniscal repairs and partial meniscectomies.¹³⁷ Meniscal repairs, which consist of repairing a tear via suturing, are typically performed only for tears located in the vascular region of the meniscus that is associated with the meniscosynovial junction.¹³⁷ Because of the specific

nature of the types of injuries that can be repaired with this technique, partial meniscectomies are more common. In the United States alone, 690,000 partial meniscectomies were performed in 2006.⁶ A partial meniscectomy consists of removing the damaged tissue from the meniscus while leaving behind as much normal meniscal tissue as possible. Because the removed tissue extends into the avascular region, the limited access to blood flow results in a low repair rate, resulting in a hole in the meniscus that leaves a loss of functionality.

Consequently, meniscal allografts and tissue-engineered implants have been researched extensively to restore functionality to the knee. Although allografts have been shown to decrease pain and improve knee function in patients in the short term, inherent shortcomings of allogeneic tissue include limited supply, potential disease transmission (such as human immunodeficiency virus or hepatitis), and risk of infection or an immunologic response. The grafts have also been shown to demonstrate some shrinkage via magnetic resonance imaging (MRI) and a reduction in mechanical strength, leading to tears and dysfunction of the allograft.¹³⁸ More recently, tissue-engineered meniscus implants have been developed as a viable alternative to allografts, as reviewed extensively by Brophy and Matava⁶ and van Tienen et al.¹³⁹ The tissue-engineered graft serves as a scaffold structure at the defect site, which allows for cell infiltration from the surrounding native tissue into the structure. The cells are then able to regenerate new tissue while the scaffold structure sustains the mechanical loading and degrades away for eventual total replacement with regenerated, functional tissue filling the defect site.

This section will focus on three synthetic options for meniscal replacement: collagen meniscus implant (CMI), hydrogels, and polymer scaffolds. These implants have all undergone extensive in vitro and in vivo testing for biocompatibility, cell response, biodegradability, and mechanical properties.

The CMI is a purified type I collagen scaffold derived from bovine Achilles tendon, and glycosaminoglycans are added to the collagen, after which the structure is cast in a mold, lyophilized, and cross-linked in formaldehyde.¹⁴⁰ The implant requires a meniscal rim and intact anterior and posterior meniscal horns for attachment during surgery and is used to treat medial and lateral meniscus injuries.^{140,141} Multiple clinical studies were conducted in which the CMI scaffold was implanted in patients with meniscal tears to evaluate the feasibility of the scaffold. In terms of the chondroprotective capabilities of the CMI, several studies have shown that the joint space and chondral surfaces were preserved after implantation.^{141–145} MRI revealed that at 5 to 10 years after surgery, new meniscal tissue was formed and integrated well with host tissue. However, the neotissue differed in MRI signal from the surrounding native tissue, suggesting that the regenerated tissue did not completely match the native tissue in structure and composition.^{144,146,147} In most cases the newly formed meniscus tissue was reduced in size compared with the host tissue, but based on patient scoring, these differences were not found to be clinically significant.^{144,146,147} In addition, the defect filling was estimated to reach 70% in 1 year.^{141,146} Mixed results on implant resorption have been reported, spanning from no observable resorption to complete implant degradation in 5 years; as such, in all cases, no inflammation or negative

effects have been reported. To assess the efficacy of the CMI compared with a control group that underwent partial meniscectomy without use of a scaffold, Rodkey et al.¹⁴⁶ followed up on 311 patients (aged 18 to 60 years) who underwent either CMI implantation or a partial meniscectomy. Results showed that patients treated with CMI had an improvement in activity levels, as evaluated at 5 years using the Tegner index for activity, whereas Lysholm scores for pain were found to be the same for both procedures. Moreover, the number of revision surgeries required was reduced by 50% with the CMI.¹⁴⁶ A 10-year study was performed by Zaffagnini et al.¹⁴³ with 36 male patients (aged 24 to 60 years). Regardless of CMI implantation, two patients per group required revision surgery during the 10-year period. The improved activity level compared with the control group at 5 years as evaluated by the Tegner index was also observed here, with a similar activity level being maintained for over 10 years. In addition, patients implanted with the CMI showed either significant pain improvement at 10 years (when evaluated using a visual analog pain scale) or a similar pain score compared with the control group (when evaluated using the Lysholm score, as was found previously by Steadman and Rodkey¹⁴¹).¹⁴³ In a recent systematic review of 11 studies (396 patients) to compare clinical outcomes and complications of CMI by Grassi et al.,¹⁴⁸ Lysholm score and visual analogue scale (VAS) showed an improvement at 6 months up to 10 years. The Tegner activity level, although peaking at 12 months and declining at subsequent evaluations, was reported to be above the preoperative level. Overall, this review concluded that CMI could produce good and stable clinical results, particularly regarding knee function and pain, with low rates of complications and reoperations.¹⁴⁸ CMI obtained FDA approval in 2015.

Another class of meniscal implants is based on polymeric materials. Actifit is an aliphatic polyurethane implant developed for medial and lateral partial meniscus tears. The graft is composed of two types of degradable polymers with distinct mechanical properties: 80% of a mechanically soft polymer poly(ϵ -caprolactone) and 20% of a mechanically stiffer segment polyurethane.¹⁴⁹ This polymer blend was optimized through *in vitro* and *in vivo* testing for mechanical properties, degradation properties, and biocompatibility.^{150–153} The scaffold exhibits a relatively slow degradation rate (it takes up to 5 years to fully degrade).¹⁵⁴ It has also been shown to improve contact area and pressure in a sheep cadaver model,¹⁵⁵ and in testing in the canine model, tissue ingrowth into the scaffold and capsule were noted at 6 months.¹⁵⁶ During clinical testing by Verdonk et al.,^{157,158} of the 52 patients with partial meniscus defects, patients receiving the Actifit implant were compared with patients undergoing a standard partial meniscectomy. For one group treated with Actifit, tissue ingrowths of 85.7% were observed at 3 months, with 12-month biopsies showing cells with meniscus-like differentiation potential. All patients receiving the Actifit implant measured significant improvements in terms of International Knee Documentation Committee (IKDC) score, Knee Injury and Osteoarthritis Outcome score, and Lysholm knee scale, showing that in 2 years, the scaffold is capable of restoring a certain level of knee functionality. A recently published midterm follow-up report of the clinical and MRI results 4 years after Actifit showed

that sustained improvement of pain scores and knee function scores compared with baseline.¹⁵⁹ Review of MRI scans did not reveal any appreciable change in articular cartilage. Although Actifit does not yet have FDA approval, it is approved for use in Europe.

Recently, results have been published on a novel resorbable polymer fiber-reinforced meniscus reconstruction scaffolds.¹⁶⁰ Using an ovine model the integrity, tensile and compressive mechanics, cell phenotypes, matrix organization and content, and protection of the articular cartilage surfaces were studied over 1 year. Fibrocartilagenous repair with both types 1 and 2 collagen were observed, with areas of matrix organization and biochemical content similar to native tissue concluding that this resorbable fiber-reinforced meniscus scaffold may support the formation of functional “neomeniscus” tissue.

Another meniscal implant with potential is the NUsurface, a polycarbonate-urethane implant reinforced circumferentially with Kevlar fibers to mimic the functional properties of the meniscus.¹⁶¹ These fibers are based on the native menisci's oriented collagen fiber network. Preliminary studies in a sheep model showed that the implant resisted wear and mild cartilage degeneration,¹⁶² although the total osteoarthritis score was not affected.

Many other promising technologies are being investigated, although most have not yet reached the clinical testing stage. For example, biodegradable polycaprolactone nanofibers have been used to mimic the native fiber alignment of collagen found in the meniscus with promising results and potentially superior long-term biocompatibility.¹⁶³ *In vivo* testing of foam polycaprolactone scaffolds revealed the formation of meniscus-like tissue with initial mechanical properties approaching those of the native meniscus.^{151,156,164} Hydrogel scaffolds (e.g., polyvinyl alcohol) for meniscus repair have also been evaluated *in vivo* and have been shown to be chondroprotective in addition to promoting meniscus-like tissue regeneration.^{153,165–167} Although these tissue-engineered implants appear to be promising, clinical applications are pending based on preclinical and clinical outcomes. The most recent meniscal therapy approaches are incorporating MSCs in their approach.^{168–171}

IMPLANTS FOR CARTILAGE AND OSTEOCHONDRAL REPAIR

Articular cartilage lines the surfaces of joints and enables near frictionless motion and load bearing. Composed of both liquid and solid phases, cartilage is a highly specialized tissue, with complex structure-function relationships.¹⁷² It is largely avascular and aneural, and consequently, it has a limited capacity for self-repair.¹⁷³ Clinical treatments of cartilage defects include joint lavage, subchondral drilling, microfracture, and osteochondral transplantation (with autografts or allografts). However, poor long-term outcomes are associated with many of these techniques because of unwanted fibrocartilage formation and inadequate graft-to-bone integration.^{174–176} Furthermore, although the use of osteochondral allograft has demonstrated positive results, risk of disease transmission and issues with graft availability, preservation, and storage remain.¹⁷⁷ Alternative techniques have been developed and studied, including the use of autologous

chondrocyte implantation (ACI), use of particulated juvenile hyaline cartilage (DeNovo, Zimmer), and chondrogenesis using stem cells. There are many scaffold options available for ACI, as well as both a two-stage and one-stage technique, which will be reviewed next.

Brittberg et al.¹⁷⁸ published the first clinical report demonstrating the promise of ACI, a two-step procedure in which autologous chondrocytes are arthroscopically harvested from a healthy cartilage donor site (typically the intercondylar notch of the knee), expanded *in vitro*, and then reintroduced to the defect site through a second open procedure. In addition, an autologous periosteum flap is harvested from the patient and used to contain the cells within the defect site. This procedure was the first FDA-approved cell-based product for cartilage repair and served to bridge the gap between previously available techniques and total joint replacement, because this procedure eliminated issues of disease transmission, as well as graft availability, preservation, and storage. Although ACI results in satisfactory clinical outcomes,¹⁷⁹ several limitations are inherent to the procedure. Of note, the necessity of two surgical procedures increases cost and leads to extended recovery times.¹⁸⁰ Furthermore, cases of cartilage hypertrophy have been reported.¹⁸¹ Finally, it is unclear if the cells recover completely from long-term monolayer culture, if they are homogeneously distributed within the repair tissue, and how many cells are actually retained within the defect site.¹⁸²

To tackle the shortcomings of the ACI procedure, second- and third-generation techniques have been developed that use matrices that eliminate the need for periosteal harvest, support and retain the chondrocytes in a 3D matrix, and enable a homogeneous distribution of the cells within the defect. The matrices that have been developed are composed of a wide variety of materials, both natural and synthetic, and have shown great promise in initial trials. Natural matrices are attractive because they can be designed to closely mimic the native cartilage matrix. Matrix-assisted chondrocyte implantation (MACI) is a second-generation ACI technique in which a porcine-derived collagen I/III bilayer is seeded with autologous expanded chondrocytes. The use of a scaffold effectively eliminates the need for periosteal harvest. Bartlett et al.¹⁸³ conducted a randomized comparison of MACI and ACI-C (ACI with a collagen I/III flap used in place of periosteum) for treatment of chondral knee defects in 91 patients and found that the two treatments resulted in clinically comparable outcomes. More recently, MACI has been compared with microfracture by Basad et al.¹⁸⁴ in a study of 60 patients with isolated cartilage defects, and it was found that MACI was significantly more effective over time than microfracture according to three different scoring systems (the Tegner index, International Cartilage Repair Society–patient, and International Cartilage Repair Society–surgeon). Zheng et al.¹⁸⁵ performed histologic analysis on a cohort of 56 MACI-treated patients and found that this technique supported chondrocyte phenotype maintenance, which was determined by aggrecan, type II collagen, and S-100 expression. After 6 months, 75% hyaline cartilage regeneration was reported. Behrens et al.¹⁸⁶ and Ebert et al.¹⁸⁷ have both performed 5-year follow-up studies for 11 and 41 patients, respectively, and found high patient satisfaction and low failure rates. MACI received FDA approval in 2016.

The Cartilage Regeneration System (CaReS) also uses a collagen matrix for cell-based cartilage regeneration; however, CaReS is made from a rat-derived collagen type I matrix that is seeded with primary cells that have not been expanded in monolayer culture. This technique is based on the assumption that cells that have not been exposed to monolayer expansion are more effective for regenerating cartilage tissue. Flohé et al.¹⁸⁸ compared the CaReS system with the MACI procedure for repair of cartilage defects in the knees of 20 patients and found that both treatments resulted in improved clinical outcomes after 1 year, with no significant differences detected between treatments. In a multicenter clinical trial, Schneider et al.¹⁸⁹ followed up with 116 patients who received the CaReS implant between 2003 and 2008. The overall treatment satisfaction was judged as good or very good in 88% of the cases by the surgeon and in 80% of the cases by the patient. These observations are highly promising, but longer-term follow-up will be necessary to determine if the CaReS system has distinct advantages over other approaches.

Monolayer expansion of autologous chondrocytes is also avoided in the NeoCart system, in which chondrocytes are cultured on scaffolds made of bovine collagen type I in custom bioreactors that mimic the conditions of the knee through varying pressure and low oxygen tension. Crawford et al.¹⁹⁰ demonstrated the clinical safety of this system in a small trial in which eight patients received the NeoCart treatment. In this study, pain scores were significantly reduced after treatment, and none of the patients experienced hypertrophy or arthrofibrosis. More recently, the NeoCart implant was compared with microfracture in a randomized trial of 30 patients, and it was found that significantly more patients treated with NeoCart responded positively to the treatment at both 6 and 12 months, with the trend continuing at the 2-year follow-up.¹⁹¹

A more complex implant, ChondroMimetic, is composed of three natural materials and is designed to closely mimic the natural cartilage environment. This implant is a dual-layer porous plug composed of collagen, glycosaminoglycans, and calcium phosphate. The scaffold can be prehydrated with sterile fluids and autologous blood. Although ChondroMimetic is an acellular, off-the-shelf product, it can be used in conjunction with ChondroCelet, which is a cell-based technology that is offered by the same company, TiGenix. ChondroMimetic was launched in October 2010 in Europe; patient enrollment in an open-label extension study just commenced in July 2017.

In addition to collagenous implants, several hyaluronan-based matrices have been developed. Hyalograft C is a hyaluronic acid scaffold (HYAFF) that is combined with autologous chondrocytes. Hyalograft C can be used in both arthroscopic and open procedures, and satisfactory clinical outcomes have been reported after 7 years.¹⁹² Improved clinical outcomes were reported for young patients in a prospective study of 70 patients with 3- and 4-year follow-up and in a study of 36 patients with 2- and 3-year follow-up analysis.^{193,194} A study of 62 patients with 7-year follow-up found that, when compared with female patients, young active men had the best clinical outcomes when treated with Hyalograft C.¹⁹² Nehrer et al.¹⁹⁵ reported that although Hyalograft C resulted in satisfactory repair for patients with a primary indication (e.g., young patients with a stable and healthy knee joint with an

isolated chondral defect), it is a poor option for salvage procedures or for patients with osteoarthritis. Kon et al.¹⁹⁶ compared Hyalograft C with microfracture in 41 professional or semiprofessional soccer players and found that although microfracture allowed players to return to competition more quickly, repair with Hyalograft C may offer more durable clinical results. In addition, Hyalograft C was compared with MACI by Kon et al.¹⁹⁷ in a trial of 61 patients who were older than 40 years. A faster improvement in the IKDC subjective score was reported for the patients treated with Hyalograft C, whereas similar scores were reported at the 2-year follow-up.

A more recent product, BioCart II, combines recombinant hyaluronan with homologous human fibrin to form a macroporous sponge that is seeded with autologous chondrocytes that have been primed with a recombinant fibroblast growth factor 2 variant. A preliminary study by Nehrer et al.¹⁹⁸ reported good defect filling with the BioCart II system in a study of eight patients. Significant improvement in defect healing over time was subsequently reported by Eshed et al.¹⁹⁹ in a study that evaluated 31 patients at time points ranging from 6 to 49 months after BioCart II implantation.

Another scaffold based on a polymer derived from nature is BST-CarGel, which is an injectable chitosan-based scaffold that is used in conjunction with bone marrow stimulation to form a volume-stable clot that drives cartilage regeneration. BST-CarGel is injected into the defect in a single-step procedure and cross-linked in situ. Shive et al.²⁰⁰ followed up with 33 patients treated with BST-CarGel and reported preliminary evidence suggesting that BST-CarGel has the potential for treatment of focal cartilage defects with varying etiology. In addition, alginate- and agarose-based scaffolds such as the Cartipatch system and a bead system have been investigated; however, few clinical reports of these approaches have been published.^{201,202}

In addition to naturally derived products, several synthetic polymer-based scaffolds are currently on the market in Europe. BioSeed-C is a polyglycolic/PLA- and polydioxane-based material that is combined with culture-expanded autologous chondrocytes that are suspended in fibrin. Kreuz et al.²⁰³ followed up on 19 patients with osteoarthritis who had received BioSeed-C treatment and reported good clinical outcomes 1 year after implantation. Moreover, BioSeed-C remained stable over the course of a period of 4 years, suggesting that it may be a promising treatment option for the repair of focal degenerative cartilage defects in the knee.²⁰³ In a larger study of 52 patients with full-thickness defects by Kreuz et al.,²⁰⁴ BioSeed-C treatment resulted in good clinical outcomes after 4 years despite a persisting deficit in mechanical strength. The authors suggest that this deficit may be addressed with a focus on muscular strength during rehabilitation.²⁰⁴

The Cartilage Autograft Implantation System (CAIS) is another polymer-based approach which consists of absorbable copolymer foam of 35% polycaprolactone and 65% PGA, reinforced with a polydioxanone (PDO) mesh (Advanced Technologies and Regenerative Medicine). In a one-step procedure, autologous cartilage is harvested, minced, and uniformly distributed within the scaffold using a fibrin sealant. The polymer foam is designed to keep the tissue fragments in place while the PDO mesh enables the foam to have adequate mechanical strength during implant

handling. Cole et al.²⁰⁵ compared CAIS treatment with microfracture in a randomized study of 29 patients at 1 and 2 weeks and periodically up to 2 years after surgery. It was found that CAIS resulted in significant increases in select subdomains in the Knee Injury and Osteoarthritis Outcome Score assessment tool, and it was concluded that CAIS is a safe, feasible, and effective method for treating patients with focal chondral defects that may improve long-term clinical outcomes.²⁰⁵

In addition to chondral grafts, several implant systems have been developed to address both chondral and osteochondral defects. Cartiva is a polyvinyl alcohol cryogel that has been used in patients since 2002 and consists of cylindrical gels that can be used to replace osteochondral grafts (either autografts or allografts). Falez and Sciarretta²⁰⁶ performed a preliminary clinical study and concluded that the use of this type of treatment should be limited to precise indications: grade 3 and 4 chondral or osteochondral symptomatic defects, focal unicompartimental defects with 15-mm maximum extent, limitation of the patient's age to the fourth to seventh decade of life, or absence of angular deformities or articular instabilities. Although cases of failure and dislocation have been reported,²⁰⁷ the synthetic cartilage resurfacing technique has the advantages of no donor defect, one short-step surgical procedure, and immediate weight-bearing ability.²⁰⁶

The TruFit CB plug is also a synthetic osteochondral implant that consists of a porous PLGA scaffold that is reinforced with PGA fibers and calcium sulfate mineral. Dhollander et al.²⁰⁸ investigated the TruFit CB for osteochondral repair and observed modest outcomes. In a more recent study, Joshi et al.²⁰⁹ reported that, although the TruFit CB system led to initial symptom relief in 10 patients with a median age of 33.3 years, a failure to regenerate subchondral bone over a 2-year period was observed, which in the long term could lead to implant failure and a repeat operation.

In a systematic review of the use of scaffolds in the repair of articular cartilage lesions by Filardo et al.,²¹⁰ nine approaches were identified using two-stage procedures and seven approaches were identified using single-stage procedures. It was noted that, although there are a multitude of cell/scaffold options available, well-designed studies exploring efficacy and long-term outcomes are lacking. Overall, a plethora of implants based on both natural and synthetic materials for cartilage and osteochondral repair are commercially available. Implants offer advantages over the ACI procedure because they circumvent the use of periosteal tissue, they address the challenge of homogenous cell distribution and retention within the defect site, and they can provide a 3D matrix that supports chondrocyte phenotype maintenance. Although most implants are engineered to regenerate cartilage tissue, osteochondral implants provide support for both bone and cartilage tissue regeneration. As mentioned previously, the next frontier in cartilage restoration will likely augment the aforementioned scaffolds with MSCs. Kon et al.²¹¹ performed a systematic review of scaffold treatments with and without the use of cells and found that the majority of studies that incorporated cells found them superior to scaffold alone (71 of 89 articles). However, given the variety in scaffold products and types of cells used, it has proven difficult to fairly compare outcomes. Looking forward, approaches may focus on the recruitment of appropriate cells into acellular

scaffolds, development of optimal scaffold microstructure, and integration of grafts with host tissue.¹⁷⁰

SUTURES

Many of the synthetic implants discussed in this chapter require fixation via suturing at the site of implantation. Similar to the implants described previously, biocompatibility and the mechanical properties of the suture must be considered. In addition to sustaining physiologic loading, suture mechanical properties also need to withstand pressure from knot-tying techniques.²¹² In this section, nonabsorbable versus absorbable sutures and monofilament versus braided sutures are discussed. Nonabsorbable sutures are fabricated from inert materials that do not degrade and are left at the repair site permanently or removed manually. Absorbable sutures degrade over time within the body; they are typically made of polymers that degrade by hydrolysis or enzymatically. Each type of suture has advantages for certain tissue repair procedures and can further be broken down into monofilament and braided sutures. Monofilament sutures are composed of a single fiber, whereas multifilament sutures are composed of multiple fibers, often encased in an outer sheath. Braided sutures are made up of multiple fibers and have the advantages of a grooved surface that prevents knots from loosening, improved handling, and low memory (i.e., a minimal tendency for the suture to recoil back to its original position). Sutures may be composed of single polymers or polymer blends, depending on the desired mechanical and degradation requirements.

Nonabsorbable sutures are made from inert materials such as nylon (e.g., Dermalon, Monosof, Surgilon, Nurodon, and Ethilon), polybutester (e.g., Novafil and Vascufil), polyester (e.g., Surgidac, Ti-cron, and Cottony II), polyethylene (e.g., MaxBraid, Mersilene, and Ethibond), and polypropylene (e.g., Surgipro, Deklene, and Prolene). These materials all elicit a low level of inflammatory response.²¹³ These nonabsorbable sutures are commonly used for deep tissue repair, where long-term support is required, or for skin closures, where they are manually removed after healing.²¹⁴ Although nonabsorbable sutures only risk causing a foreign body response upon implantation, absorbable sutures may elicit further responses from the body during degradation because of degradation products in the local environment.

Absorbable sutures are fabricated from biodegradable polymers. Although many sutures have been developed from materials that elicit minimal inflammatory responses during degradation, a few materials that are associated with moderate foreign responses are still commercially available. Common biodegradable polymers used in sutures include PGA (e.g., Dexon S), PLLA, PDO (e.g., Monodek and Ethicon), and poly-D, L-lactic acid and their copolymers.²¹² These materials exhibit varied rates of degradation, with a higher lactic acid content corresponding to a slower resorption rate.²¹² In terms of inflammatory response, studies have shown that polyglyconate (e.g., Maxon) and PDO sutures elicit less of a foreign body response than do materials such as PGA (e.g., Dexon) and polyglactin (e.g., Vicryl).^{214–219} Cat gut and silk fibers have been shown to promote a moderate to intense inflammatory response and are commonly used for skin.²¹² Other examples of commercially absorbable sutures,

which can include modifications that serve different purposes for varying knot-tying properties and tensile strengths, are PGA with a polycaprolactone:PGA coating (e.g., Bondek Plus), poliglecaprone 25 (e.g., Monocryl), polycaprolate (e.g., Dexon II), PDO with polyglactin 910 coating (e.g., Vicryl), and polyglytone (e.g., Caprosyn).

Monofilament sutures minimize foreign body response because of the low surface area that is exposed to the body.^{220,221} This reduction in surface area also results in a lower hydrolysis rate, and thus these sutures maintain their mechanical properties longer than other sutures.^{218,219} They are also more easily removed but have issues with high memory, which complicates handling of the suture during surgery because of the stiffness of the material, causing the suture to recoil back to its original shape. Braided sutures have unique advantages related to knot-tying capabilities.^{222–225} The braided structure helps the suture to maintain the knot structure and minimizes the risk that the suture will come undone after surgery. Braided sutures also have low memory, making them much easier to handle than monofilament sutures. Further modifications have been made to these sutures to improve their knot-tying capabilities. For example, Ethibond is a braided polyester suture coated with polybutylate for a slicker surface that improves arthroscopic knot tying.^{226,227} Tevdek and Polydek have PTFE coatings to improve their strength, whereas FiberWire was developed to combat the issue of suture breakage during arthroscopic knot tying. It is composed of ultra-high-molecular-weight polyethylene (UHMWPE) surrounded by a polyester braid.^{213,214,225,228,229} Braided sutures composed entirely of UHMWPE have been developed more recently (e.g., ForceFiber, MagnumWire, Ultrabraid, and Hi-Fi). Orthocord is another UHMWPE suture with a PDO core, and it is coated with polyglactin 910, which makes the surface of the suture more frictionless to improve knot-tying characteristics. Thicker material such as FiberTape (Athrex) have been developed, and in studies compared with No. 2 suture in double row rotator cuff repair, the thicker FiberTape has shown threefold higher contact pressures at the footprint and 1.5 times higher load to failure.²³⁰ However, despite these superior biomechanical properties, the difference in retear rates at 6 months postoperative (16% in the tape group and 17% in the suture group) was not statistically significant.²³⁰

Despite the benefits of the braided sutures for improved handling, the grooved formation introduces complications. Contamination is more prominent in braided sutures because of the structure of the fibers, specifically in the grooved areas where immunocompetent cells are unable to infiltrate.^{223,224} Thus many sutures with antimicrobial coatings have been developed, with an example being the triclosan coating from Ethicon on the Vicryl, PDO, and Monocryl sutures.^{222,231–234} The triclosan coating was found to inhibit the growth of the bacterium *Staphylococcus aureus* in vitro and in vivo in a guinea pig model over 48 hours.

Recent laboratory work has extended the function of sutures for growth factor delivery and improved mechanics. A number of studies have demonstrated the potential of sutures to deliver growth factors to the repair site for enhanced healing. For example, animal model work showed that the growth factor PDGF-BB, delivered from suture surfaces, improved tendon healing.^{235,236} More recently, porous sutures were developed, allowing for

dramatically higher levels of growth factor delivery without compromising the mechanical properties of the sutures.²³⁷ To improve load transfer between sutures and the tissue they are grasping, adhesive-coated sutures were also developed.²³⁸ In an example case of flexor tendon repair, such an approach could improve the repair strength by twelvefold.

SUTURE ANCHORS

Different sutures are used in combination with a variety of suture anchors, depending on the needs of the damaged tissue to be repaired. Suture anchors are widely used in orthopedic surgery, especially in shoulder procedures, such as rotator cuff^{239,240} and glenoid labrum repair,^{154,241} as well as in the elbow.^{242,243} A typical suture anchor consists of the anchor, which is placed in bone during surgery, and its suture, which comes either preloaded by the manufacturer or loaded at the time of surgery through an eyelet on the anchor. Most sutures used in suture anchors are UHMWPE, which have a higher resistance to breakage but fail through slippage, making knot tying critical.^{244,245} This section will focus on the design and performance of the anchor.

A typical suture anchor resembles a screw with an eyelet loaded with suture material. Details of the design parameters, such as the length and diameter of the screw, the number of threads on the screw, and the location of the eyelet on the screw, depend on the properties of the bone tissue where the anchor will be placed to reattach soft tissue to bone. For rotator cuff repair, the greater tuberosity of the humeral head is the targeted region for placement of suture anchors. In general, the cortical bone of the greater tuberosity is thin, and thus excessive decorations during bone surface preparation in surgery could compromise the pullout strength of anchors that rely on cortical or subcortical fixation. In addition, trabecular bone density of the greater tuberosity varies with location and pathologic states.^{231,246} For example, in a torn rotator cuff, the bone density of the greater tuberosity decreases significantly because of decreased mechanical loading in the region.^{110,212,247} Because the bone mineral density can affect anchor fixation and pullout strength, this factor should be considered when choosing a suture anchor.

Suture anchors can be divided into screw anchors and impaction anchors based on the way in which they are fixed to bone. As its name suggests, the screw anchors are threaded, which helps them to advance into bone and hold the anchor in place once implanted. The screw anchor has a major and a minor diameter; the major diameter is the entire width of the anchor, and the minor diameter is the width of the inner core of the anchor. On the other hand, impaction anchors are threadless. These anchors are placed into the bone through external impaction onto a cortical hole that has a slightly smaller diameter compared with that of the anchor. After placement in the bone, the anchor expands, which prevents it from pulling out. Compared with impaction anchors of similar size, threaded anchors displace less bone but provide an improved holding strength because of the increased contact surface area between anchor and bone, given the presence of threads.^{248–250}

Based on how the sutures are tightened after reattaching soft tissue to the bone, suture anchors can be categorized as either

normal or knotless anchors. Knotless anchors have a special eyelet-suture system that makes suture tying unnecessary. The knotless anchor design has become instrumental in the lateral row of double-row rotator cuff repair.^{251–254}

Suture anchors can also be categorized on the basis of their material. Early suture anchors used for shoulder surgery were made with metals (primarily titanium and stainless steel). Before suture anchors came into use, metals were already widely used in orthopedic implants such as total hip prostheses and had been shown to provide rigid fixation. Metal anchors can be visualized on standard postoperative radiographs, which makes it possible to assess anchor migration and potential treatment failure. These anchors are also bioinert and have minimal osseous integration with the host bone tissue. For example, it was found that stainless steel anchors became encapsulated by a fibrous layer, whereas titanium anchors induced a minimal inflammatory response. However, obvious disadvantages were also associated with the metal anchors. The first disadvantage relates to postoperative imaging. MRI is the preferred imaging modality for evaluating glenohumeral pathology, but metal anchors can cause distortion of the scans. The second disadvantage is that if a revision surgery is required, the previously placed anchors could make placement of new anchors difficult because the former may not be easy to remove. In addition, complications with metal anchors, such as chondral damage caused by anchor loosening and fatigue, have been reported. Therefore demand remains for alternative materials for suture anchors.

To avoid compromising the anchor performance criteria while satisfying the need for an anchor that does not interfere with imaging or additional surgical procedures, bioabsorbable anchors were developed. These anchors are more biocompatible than metal anchors but provide the same initial fixation strength of soft tissue to bone. However, biodegradable anchors should be absorbed at appropriate rates (i.e., not so fast that the anchor degrades and fails to maintain its mechanical strength before the newly formed tissue regains mechanical integrity, but not so slow that the anchor stress shields the repair site and prevents healing). Furthermore, the bioabsorbable anchors should be made from biocompatible materials that elicit minimal toxicity, antigenicity, pyrogenicity, or carcinogenicity. Given these requirements, polymers have become the preferred material for suture anchors. Early bioabsorbable suture anchors were made from PGA because it was already used in other FDA-approved biomedical devices. However, its relatively fast degradation profile (3 to 4 months) was reported to be associated with early loss of fixation, osteolysis, loose body formation, and glenohumeral synovitis. Therefore PLA, which has a much slower degradation profile, was used in the next generation of suture anchors. Currently, most bioabsorbable suture anchors are made from PLA, copolymer of PLA and PGA (PLGA), or a combination of the two. Although they are developed to outperform metal anchors, bioabsorbable anchors still could lead to postoperative complications such as bone osteolysis, chondral damage, and significant patient morbidity. To bridge the gap from initial fixation strength to eventual bone formation at the anchor placement site, without inducing osteolysis or synovitis, biocomposite anchors were then developed. These anchors

consist of a bioabsorbable polymer and an osteoconductive bioceramic such as β -tricalcium phosphate. As the anchor degrades, so does the ceramic, resulting in the release of calcium and phosphate substrates, providing an environment that promotes bone formation. Compared with anchors that are made from solely from bioabsorbable polymers, biocomposite anchors may accelerate new bone formation. As a result, few inflammatory reactions and complications have been reported with the use of these anchors.

Polyetheretherketone (PEEK) has been developed as an anchor material.²⁴⁶ It is a nonbiodegradable, bioinert polymer that is radiolucent and resists chemical, thermal, and radiation-induced degradation. Because of these properties, suture anchors made from PEEK provide high initial repair strength, minimal bone ingrowth, minimal inflammatory responses, and good postoperative imaging. In addition, because it is a plastic, PEEK is soft enough to be drilled through, which makes revision surgeries possible. Thus although they were introduced relatively late compared with metal and biodegradable anchors, PEEK anchors have quickly gained popularity in the market.

Because of the functional requirements, the most important parameter used to evaluate suture anchor performance is pullout strength. Barber et al.^{79,110,246,255-257} has routinely evaluated and compared the pullout strength of commercially available suture anchors on porcine femurs. Previously they used a single-pull destructive test to discern pullout strength^{246,255,258,259}; however, more recent studies use cyclic loading to more realistically stimulate the postoperative setting.^{260,261} In addition, the glenoid anchors are typically placed in dense cortical bone, whereas anchors for rotator cuff repair are placed in cancellous bone. In previous studies from 2003 through 2011, 66 types of commercial suture anchors were tested, and it was determined that the main failure mode of metal anchors was suture breakage, the main failure mode of bioabsorbable anchors was eyelet breakage, and for PEEK anchors, failure by anchor pullout was also observed in addition to eyelet breakage.^{57,246,255,258,259} In the 2013 study by Barber et al., strong UHMWPE suture was used and suture breakage was not a factor.²⁵⁶ Twenty-four cuff anchors and 13 glenoid anchors were trialed. Each anchor was tested 20 times, 10 samples in cortical bone and 10 in cancellous. All were loaded with UHMWPE suture. The samples underwent 200 cycles of a 10- N to 100-N load. After the 200 cycles were performed, destructive testing was performed. No difference was recorded for the rotator cuff anchors or the glenoid anchors placed in cortical versus cancellous bone. The rotator cuff anchors withstood greater loads than glenoid anchors. The rotator cuff anchors failed mostly by eyelet breakage, whereas the glenoid anchors failed most often by anchor pullout. Of note, the all-suture anchors that were tested performed comparably with the solid-body anchors.²⁵⁶ In a study by Goschka et al.²⁵⁴ comparing all-suture anchors in double-row rotator cuff repair to solid-body anchors, the biomechanical performance under cyclic loading was comparable.

A recent study by Postl et al.²⁶² showed that augmentation of suture anchors with bioabsorbable osteoconductive fiber-reinforced calcium phosphate such as PMMA enhanced failure load by 66.8%. This may be a useful tool in patients with significant subchondral cyst formation or osteoporosis.

All-suture anchors were recently developed for rotator cuff repair to decrease bone damage. A recent study by Nagra et al.²⁶³ compared four commercially available all-suture anchors to traditional suture anchors with a cyclic loading model. The tensile strength was significantly higher for traditional anchors (181 N) compared with the all-suture anchor (133 N). The all-suture anchors failed predominantly by pullout, whereas the traditional failed by suture breakage.²⁶³ However, of note, knotless suture tape repair of a quadriceps tendon showed biomechanical superiority when compared to transosseous tunnels or suture anchor repair.²⁶⁴

Over the years, the suture anchor has evolved to become an instrumental tool in orthopedic surgery, especially in arthroscopic rotator cuff and labral repair. As its design has improved and diversified, its utility has been realized in other surgical sites such as the elbow and patella. As suture strength continues to improve and advances in knotless suture anchor strength and all-suture anchors develop, optimized surgical technique and subsequent improvement in patient outcomes will be achieved.

SUMMARY

In this chapter, current synthetic and tissue-engineered graft repair options for soft tissues associated with sports injuries were reviewed, specifically for tendons, ligaments, menisci, and cartilage. Development and implementation of these implants have reached various stages of commercialization and clinical use. In the case of tendon repair, ECM-derived tendon grafts and augmentation devices have exhibited mixed outcomes in clinical studies. Alternative treatment options such as synthetic and tissue-engineered grafts are being developed, with promising results for tendon regeneration and tendon-to-bone integration. However, several challenges remain to be overcome before the widespread clinical application of the tissue-engineered tendon implants can be realized. For ACL injuries, autografts are still the gold standard of care because synthetic and tissue-engineered grafts have yet to demonstrate their long-term efficacy and are at the in vitro or in vivo evaluation stages, but new techniques for ACL repair with collagen scaffold augmentation show promising results in early clinical trials. CMI was recently FDA approved for use in meniscal implant, and many other promising technologies such as biodegradable polycaprolactone nanofibers, foam polycaprolactone scaffolds, and hydrogel scaffolds for meniscus repair are being investigated, although most have not yet reached the clinical testing stage. The new frontier in meniscal therapy is incorporating MSCs. Finally, for cartilage and osteochondral repair, a wide variety of commercially available synthetic and tissue-engineered implants are available in Europe. In 2016 the FDA approved MACI as the first autologous expanded chondrocyte augmented scaffold for cartilage restoration. In summary, given the ever-increasing knowledge base regarding the mechanisms of musculoskeletal tissue formation and repair, as well as the promising results from the many preclinical and clinical studies currently under way, it is anticipated that both synthetic and tissue-engineered grafts will be widely used clinically for the treatment of debilitating orthopaedic injuries and

will ultimately help to improve the quality of life for numerous patients.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Barber FA, Herbert MA, Coons DA. Tendon augmentation grafts: biomechanical failure loads and failure patterns. *Arthroscopy*. 2006;22:534–538.

Level of Evidence:

II

Summary:

The purpose of this study was to determine the load to failure strengths and modes of failure of various commercially available tendon augmentation xenografts and allografts. GraftJacket, CuffPatch, Restore, Permacol, and TissueMend were tested. Failure modes differed significantly among the implant types. Understanding of the unique properties of each graft and suture placement requirements was the focus of this work rather than determining of superiority of one graft over another.

Citation:

Lu HH, Thomopoulos S. Functional attachment of soft tissues to bone: development, healing, and tissue engineering. *Annu Rev Biomed Eng*. 2013;15:201–226.

Level of Evidence:

IV, systematic review of Level I to IV studies

Summary:

This review describes the developmental processes and structure-function relationships between soft tissue and bone. It discusses the interface healing response, with a focus on the influence of mechanical loading and the role of cell-cell interactions. It also explores the current efforts in interface tissue engineering, highlighting key strategies for the regeneration of the soft tissue-to-bone interface, and concludes with a summary of challenges and future directions.

Citation:

Perrone GS, Proffen BL, Kiapour AM, et al. Bench-to-bedside: bridge-enhanced anterior cruciate ligament repair. *J Orthop Res*. 2017;doi:10.1002/jor.23632.

Level of Evidence:

III

Summary:

This review details the pathway of how a tissue-engineering strategy can be used to improve the healing of the anterior cruciate ligament (ACL) in preclinical studies and then translated to patients in an FDA-approved clinical study. It outlines the clinical importance of ACL injuries, history of primary repair, the pathology behind failure of the ACL to heal, preclinical studies, the FDA approval process for a high risk medical device, and the preliminary results from a first-in-human study.

Citation:

Grassi A, Zaffagnini S, Marcheggiani Muccioli GM, et al. Clinical outcomes and complications of a collagen meniscus implant: a systematic review. *Int Orthop*. 2014;38:1945–1953.

Level of Evidence:

IV, systematic review of Level I to IV studies

Summary:

This systematic review summarizes and evaluates the clinical outcomes of the collagen meniscus implant (CMI) and its complication and failure rates. Eleven studies were included, the pooled number of patients involved in CMI surgery were 396. The Lysholm score and VAS for pain showed an improvement at 6 months up to 10 years. The Tegner activity level reached its peak at 12 months after surgery and showed a progressive decrease through 5 and 10 years post CMI implantation, however always remaining above the preoperative level. Only a few knees were rated as “nearly abnormal” or “abnormal” at International Knee Documentation Committee grading at all follow-up evaluations. The authors concluded that CMI had good outcomes regarding knee function and pain, with low rates of complications and reoperations.

Citation:

LaPrade RF, Dragoo JL, Koh JL, et al. AAOS Research Symposium updates and consensus. *J Am Acad Orthop Surg*. 2016;24:e62–e78.

Level of Evidence:

IV, systematic review of Level I to IV studies

Summary:

The American Academy of Orthopaedic Surgeons hosted a research symposium in November 2015 to review the current state-of-the-art biologic treatments of articular cartilage, muscle, tendon, and bone injuries and identify knowledge gaps related to these emerging treatments. This review outlines the findings of the symposium and summarizes the consensus reached on how best to advance research on biologic treatment of orthopedic injuries.

Citation:

Filardo G, Kon E, Roffi A, et al. Scaffold-based repair for cartilage healing: a systematic review and technical note. *Arthroscopy*. 2013;29:174–186.

Level of Evidence:

IV, systematic review of Level I to IV studies

Summary:

This systematic review explored the treatment of chondral and osteochondral knee lesions through the use of scaffolds, by showing surgical options and results of this scaffold-based repair approach for the healing of the articular surface. The authors selected 51 articles, with 40 focusing on two-step procedures and 11 focusing on one-step procedures.

Citation:

Barber FA, Herbert MA. Cyclic loading biomechanical analysis of the pullout strengths of rotator cuff and glenoid anchors: 2013 update. *Arthroscopy*. 2013;29:832–844.

Level of Evidence:

II

Summary:

The purpose of this study was to evaluate the biomechanical and design characteristics of suture anchors under cyclic loading. More than 30 anchors were tested, including all-suture anchors and knotless anchors. Rotator cuff anchors showed greater failure

loads than did glenoid anchors in metaphyseal bone and cancellous bone. Rotator cuff anchors failed principally by eyelet breaking, whereas glenoid anchors failed more often by anchor pullout than by any other mode. No differences in stiffness were observed across the different rotator cuff and glenoid anchors tested.

Citation:

Lu HH, Subramony SD, Boushell MK, et al. Tissue engineering strategies for the regeneration of orthopedic interfaces. *Ann Biomed Eng.* 2010;38(6):2142–2154.

Level of Evidence:

II

Summary:

One of the major challenges associated with tissue-engineered bone and soft tissue grafts is biologic fixation at the implant site. This article reviews current tissue engineering strategies for the regeneration of ligament-to-bone, tendon-to-bone, and cartilage-to-bone interfaces, which are critical for improving biologic fixation and enabling integrative repair of soft tissue with bone.

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